Udemy ML, DL, NLP
course
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Sec-33 Decision Tree
Classifier & Regressor

Lec - 3 Info Gain

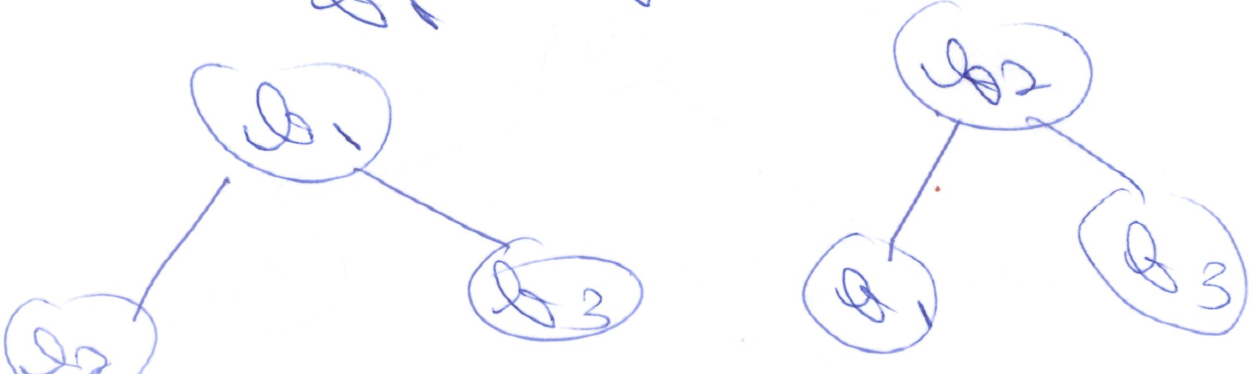
Lec - 4 Entropy vs
~~Gini~~ Gini
Impurity

Lec - 5 :- Decision Tree
split

Lec - 6 Post pruning &
Pre pruning

Information Gain.

f_1 f_2 f_3 i/p



$\text{Gain}(S, A_1)$

$$= H(S) - \sum_{V \in Val} \frac{|S_v|}{|S|} H(S_v)$$

$$H(S) = -p_+ \log_2 p_+ - p_- \log_2 p_-$$

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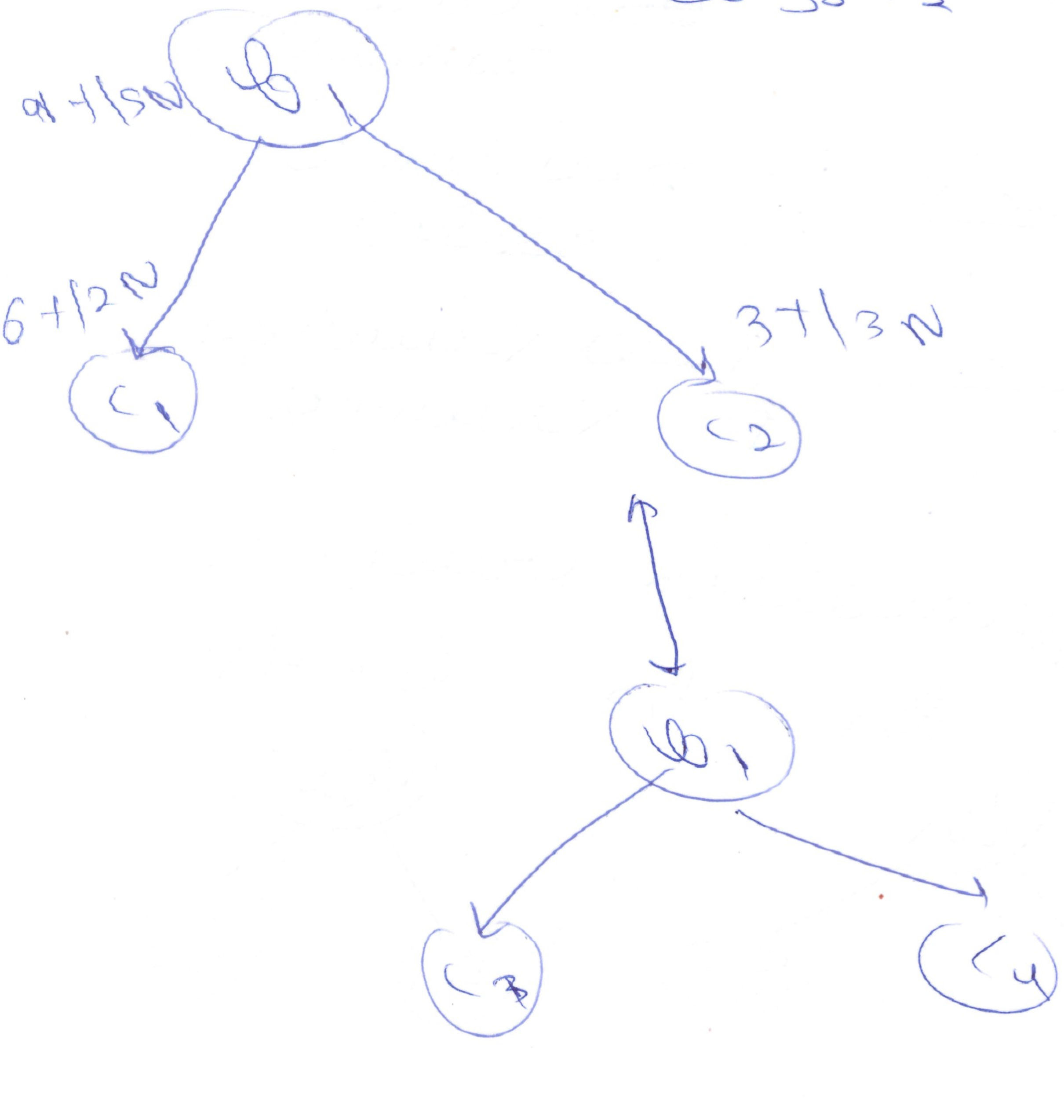


$$\Rightarrow -\frac{9}{10}\log_2\frac{9}{10} - \frac{5}{14}\log_2\frac{5}{14}$$

$$\approx 0.94$$

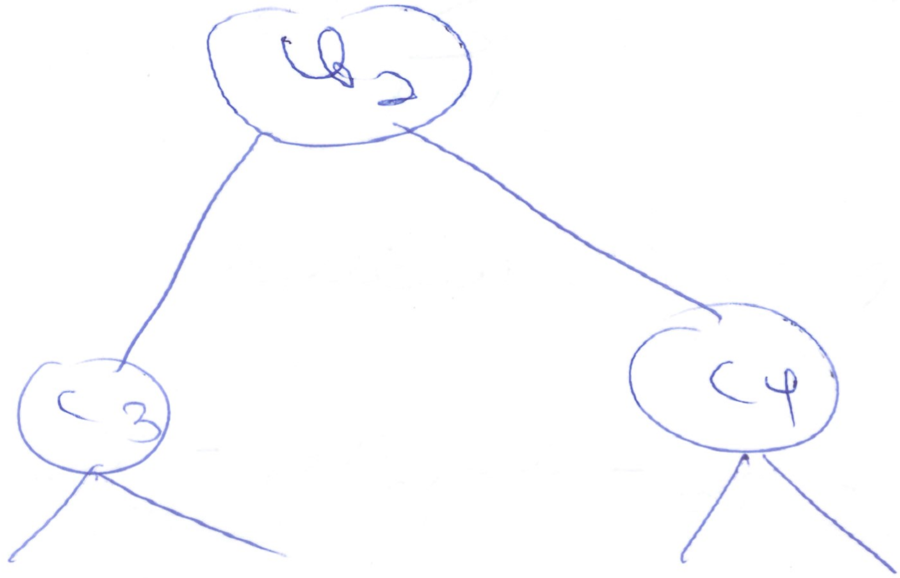
$$H(C_1) = -\frac{6}{8}\log_2\frac{6}{8} - \frac{2}{8}\log_2\frac{2}{8}$$

$$H(C_1) = 0.81$$

$$H(C_2) = 1$$

$$\text{Gain}(S, a_1) = 0.94 - \left[\frac{8}{14}\times 0.81 + \frac{6}{14}\times 1\right]$$

$$\text{Gain}(S, a_1) = 0.049$$



$$\text{Gain}(S, a_2) = 0.051$$

$$\longrightarrow \text{Gain}(S, a_1) = 0.049$$

Information is Basically calculated:

=> Entropy and Gini Impurity

Entropy vs Gini Impurity

$H(s) = -p_+ \log_2 p_+ - p_- \log_2 p_-$

O/p = 3 categories.

$GI = 1 - \sum_{i=1}^{n} (p)^2$

O/p = 3 classes

$H(s) = -p_{c_1} \log_2 p_{c_1} - p_{c_2} \log_3 p_{c_2} - p_{c_3} \log_2 p_{c_3}$

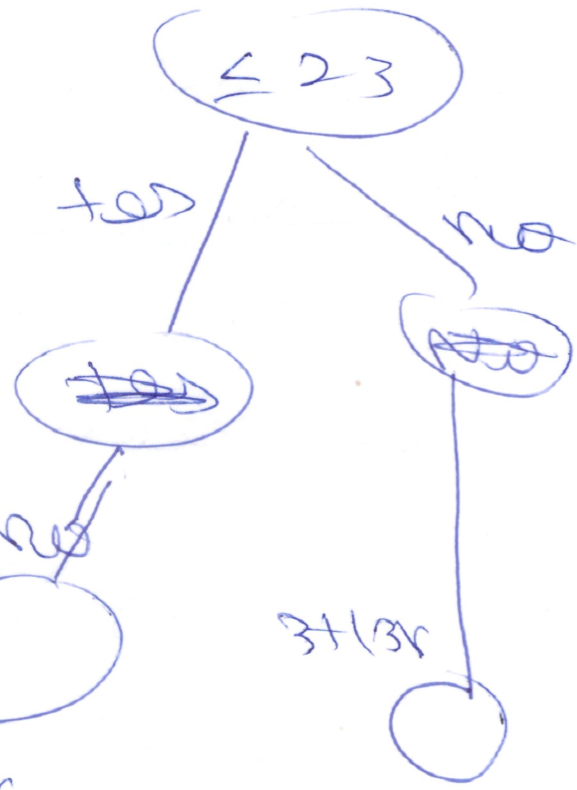
When dataset is small → Entropy

=> Decision tree for Numerical Problem

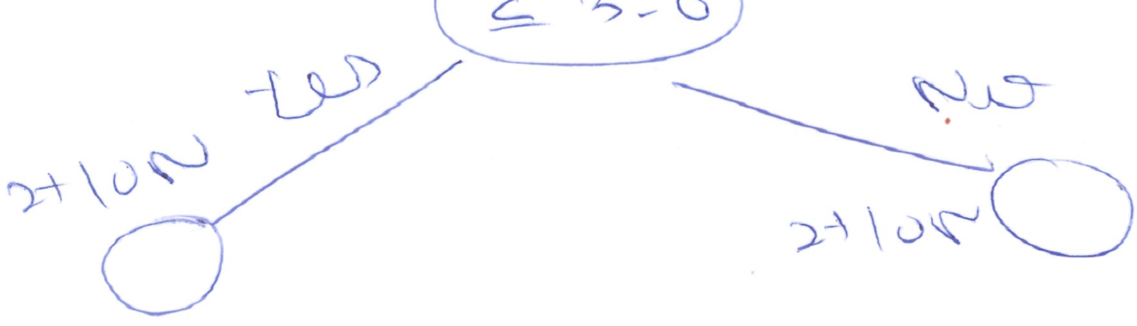
Sorting the feature values as:-

F1	O/P
2.3	Yes
3.6	Yes
4	No
5.2	No
6.7	Yes
8.9	No
10.5	Yes

(1) Threshold = 2.3

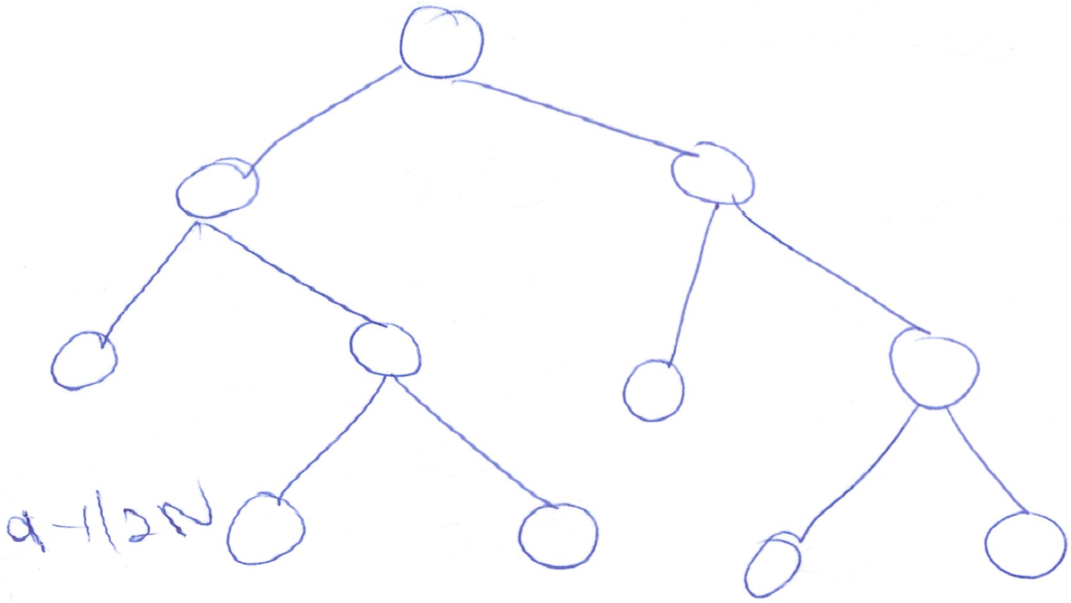


2) Threshold = 3.6



=> Post pruning and pre pruning

training dataset



→ These are the post & pre pruning decision

overfitting

train acc ↑

test accuracy ↓

(7)

Pruning:-

↳ Pruning reduces overfitting

Post Pruning:-

(1) Constructs the DT

(2) Prune it with respect to depth

(3) Smallest dataset

Pre Pruning:-

↳ Hyperparameter tuning while constructing DT.

max_depth = 3